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OF ICELAND**

Meteorological Conditions and Rural Injury Accidents in Iceland

Analysis of Accident Frequency and Traffic, 2007–2025

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FACULTY OF INDUSTRIAL ENGINEERING, MECHANICAL ENGINEERING AND COMPUTER SCIENCE

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in Software Engineering

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Analysis of Accident Frequency and Traffic, 2007–2025

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30 ECTS thesis submitted in partial fulfillment of a
Master of Science degree in Software Engineering

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Abstract

This study examines how weather conditions are associated with rural injury accidents in Iceland from 2007–2025. The main focus is on mean wind speed, with additional analyses of wind gusts, temperature, season, time of day, and daylight. Accident records were linked to 10-minute weather observations from nearby stations. The number of accidents under different weather conditions was then compared with how often those conditions occurred locally. Traffic data were used in supporting analyses.

A total of 6,259 injury accidents were matched to a valid wind observation within 20 km and five minutes of the accident time. At mean wind speeds of 20 m/s or more, 68 injury accidents were observed, compared with 31.6 expected from the local frequency of such winds. This gives an observed-to-expected ratio of 2.15. For serious or fatal accidents, the corresponding ratio was 1.87, based on 12 observed accidents and 6.4 expected. High wind gusts showed a similar pattern. Analyses using traffic data pointed in the same direction, although they were based on smaller samples. The results for temperature were less consistent.

Overall, rural injury accidents occurred more often during strong wind than would be expected from how often strong wind occurred locally. There was no clear evidence that strong wind was associated with greater accident severity.

These results show statistical associations, not causal effects. Because hourly traffic data are not available at the required spatial and temporal resolution, precise accident risk per vehicle cannot be estimated for the full study period.

Ágrip

Markmið rannsóknarinnar var að kanna hvernig veðurskilyrði tengjast umferðarslysum með meiðslum í dreifbýli á Íslandi á árunum 2007–2025. Megináhersla var lögð á meðalvind, en einnig voru skoðaðar vindhviður, hitastig, árstíð, tími dags og dagsbirta. Slysagögn voru tengd við tíu mínútna veðurmælingar frá nálægum veðurstöðvum og fjöldi slysa við tiltekin veðurskilyrði borinn saman við hversu algeng þau voru á sama stað og árstíma. Umferðargögn voru notuð í viðbótagreiningu til að kanna hvort sama mynstur héldist með tilliti til umferðar.

Alls voru 6.259 slys með meiðslum þöruð við gilda vindmælingu í allt að 20 km fjarlægð og innan fimm mínútna frá slysatíma. Þegar meðalvindur var 20 m/s eða meiri urðu 68 slys, en miðað við staðbundna tíðni slíkra vindskilyrða hefði mátt búast við 31,6 slysum. Hlutfall skráðs og vænts fjölda slysa var því 2,15. Fyrir alvarleg slys og banaslys var sama hlutfall 1,87, byggt á 12 slysum á móti 6,4 væntum. Sambærilegt mynstur kom fram við háar vindhviður. Greiningar með umferðargögnum sýndu einnig sömu fylgni, þótt þær byggðu á minna úrtaki. Tengsl hitastigs við slys voru óreglulegri og mismunandi eftir greiningaraðferð.

Ætla má að slys með meiðslum verði hlutfallslega oftar í miklum vindi, út frá því hve oft slíkur vindur mælist. Ekki eru þó skýrar vísbendingar um að mikill vindur tengdist meiri alvarleika slysa. Rannsóknin sýnir tölfræðileg tengsl en ekki orsakasamband. Þar sem umferðargögn á klukkustund eru ótiltæk, er ekki hægt að meta nákvæma slysaáhættu á hvert ökutæki.

Contents

Abstract	v
Ágrip	v
Abbreviations	xiii
1 Introduction	1
1.1 General	1
1.2 Aim and Research Questions	1
1.3 Thesis Outline	2
2 Background	3
2.1 Wind and Rural Road Accidents	3
2.2 Temperature, Daylight, and Time of Day	3
2.3 Traffic and Accident Rates	4
3 Data and Methods	5
3.1 Data	5
3.1.1 Source Data	5
3.1.2 Accident Data	6
3.1.3 Weather Data	6
3.1.4 Traffic Data	8
3.2 Data Preparation and Reproducibility	9
3.3 Analysis	10
3.3.1 Weather-Frequency Comparison	10
3.3.2 Matched-Time Comparison	11
3.3.3 Use of Traffic Data	12
3.3.4 Road-Section Traffic Comparison	12
3.3.5 Daily Traffic Comparison	13
4 Results	15
4.1 Study Sample and Coverage	15
4.2 Accident Frequency by Weather	16
4.2.1 Mean Wind	16
4.2.2 Wind Gust	16
4.2.3 Temperature	16
4.3 Matched-Time Results	19
4.4 Daylight, Season, and Severity	20
4.5 Traffic Results	20
4.5.1 Broader Annual-Traffic Comparison	21
4.5.2 Daily-Counter Checks	21

5	Discussion	23
5.1	Interpretation	23
5.2	Strengths	23
5.3	Limitations	24
5.4	Future Work	24
6	Conclusion	25
A	Supporting Checks	29
A.1	Adjusted Severity Model	29
A.2	Daily-Counter Sample	29
A.3	Traffic Sensitivity Checks	31

List of Figures

3.1	Locations of the 6,414 rural injury accidents. Accidents matched to mean wind of at least 15 m/s are highlighted. The map shows the spatial coverage of the study and is not a regional accident-rate map.	7
4.1	Annual rural injury-accident counts and the percentage matched to wind or temperature within 20 km and five minutes. The wind and temperature match counts are identical in every year; no missing temperature value is imputed.	15
4.2	All-year and seasonal mean-wind O/E, 2007–2025. Blue bars include all accidents with injury ($meiðsli \leq 3$); orange bars show the serious-or-fatal subset ($meiðsli \leq 2$). Expected counts use local weather frequency; numbers above bars are observed accidents.	17
4.3	All-year and seasonal wind-gust O/E. Gust is taken from the ten-minute observation matched to the accident time; it is not a daily maximum. Orange bars are a subset of the blue bars; numbers above bars are observed accidents.	18
4.4	All-year and seasonal temperature O/E, standardised by matched temperature station and study season. Serious-or-fatal accidents are a subset of all injury accidents; numbers above bars are observed accidents. The non-monotonic pattern is interpreted cautiously.	19
A.1	Geographic coverage of located daily counters and rural injury accidents retained or excluded by the daily-counter linkage, 2019–2024.	30

List of Tables

3.1	Source datasets. The study period is 2007–2025.	5
3.2	Definitions used throughout the thesis	6
3.3	Share of accidents with weather information from a station with a valid measurement within 20 km. Distance is shown as median / P90; every retained observation is within five minutes of the accident time.	8
3.4	Wind-data scope and quality audit, 2007–2025. Quality-rule shares use all supplied observations. Every supplied station-year contains at least one wind measurement.	8
3.5	Official traffic averages and the spring/autumn daily traffic value used in the annual-traffic analysis.	9
3.6	Data preparation and analysis workflow	10
4.1	Accident selection and analysis sample sizes.	15
4.2	Traffic inputs for the supporting vehicle-kilometre analyses.	21
4.3	Estimated rural injury-accident rate by mean wind speed, all traffic periods	21
4.4	Main result and supporting comparisons.	22
A.1	Adjusted odds ratios for a serious-or-fatal outcome among recorded injury accidents. All listed variables are included in the same model.	29
A.2	Selection of accidents for the observed daily-counter analysis	30
A.3	Characteristics of accidents retained and excluded by the daily-counter linkage	30
A.4	Upper daily-counter estimate under three assignment distances	31
A.5	Comparison and quality checks for the traffic analyses	31

Abbreviations

ADU	Average annual daily traffic.
SDU	Summer daily traffic, June–September.
VDU	Winter daily traffic, December–March.
VHDU	Spring/autumn daily traffic derived from ADU, SDU and VDU.
O/E	Observed accidents divided by expected accidents.

1 Introduction

1.1 General

Weather affects both driving conditions and the decision to travel. Strong wind can make a vehicle harder to control, low temperature can coincide with slippery roads, and darkness can reduce visibility. These conditions are important in Iceland, where weather varies greatly across locations and over short periods.

This thesis studies rural accidents involving injury in Iceland from 2007–2025. Accident times and locations are linked to nearby ten-minute weather observations. The main comparison concerns mean wind speed. Wind gusts, temperature, season, time of day, daylight, and the available traffic data are then examined separately.

Accident counts alone do not measure accident risk. Severe weather can increase the hazard faced by each vehicle while reducing the number of vehicles on the road. Weather frequency and traffic volume are therefore considered separately Andrey et al. (2003); Lord and Mannering (2010); Washington et al. (2020).

1.2 Aim and Research Questions

The aim is to quantify how the occurrence of rural injury accidents varies with weather and what the available traffic data add to the mean-wind result. Three questions guide the analysis:

1. How do observed/expected ratios for all injury accidents and their serious-or-fatal subset vary across mean-wind, wind-gust, and temperature intervals during the year and within each season?
2. Are the weather-frequency results supported when each accident time is compared with matched non-accident times?
3. How do weather-related accident rates change after including the available annual and daily traffic data?

The primary 2007–2025 analysis addresses the first question, the matched-time analysis the second, and the smaller traffic-linked samples the third. Serious-or-fatal accidents are a subset of all injury accidents. The study estimates associations, not causal effects or a complete national accident rate per vehicle-kilometre.

1.3 Thesis Outline

Chapter 2 gives the background for the weather and traffic comparisons. Chapter 3 describes the data, data preparation, and statistical methods. Chapter 4 presents the results, beginning with mean wind. Chapter 5 discusses their interpretation and limitations, and Chapter 6 gives the conclusions. The appendix contains selected checks that support interpretation of the main results.

2 Background

2.1 Wind and Rural Road Accidents

Crosswind can produce overturning, lateral sideslip, or excessive yaw. The critical condition depends on vehicle geometry and mass, vehicle speed, road friction, wind direction, and driver response rather than wind speed alone Baker (1986, 1994). High-sided vehicles are particularly exposed, but the physical mechanisms also provide a reason to examine loss-of-control accidents more generally. These engineering studies establish physical plausibility and vehicle-specific failure mechanisms; they do not by themselves establish a population-level association between wind and accident occurrence.

This problem is directly relevant in Iceland, where terrain can accelerate and redirect near-surface wind over short distances. An Icelandic reliability study modelled accident probability as the combined outcome of wind velocity and direction, road camber and friction, and vehicle speed Snæbjörnsson et al. (2007). That work explains why a weather station is an imperfect proxy for conditions at a specific road location and why results should be compared under several distance limits. Mobile road measurement research similarly shows that roadside terrain creates marked spatial variation that a sparse fixed-station network cannot fully resolve Pu et al. (2018).

Vegagerðin has developed a vehicle-wind model to support decisions about hazardous conditions and road closures, while also noting the limited experimental evidence available for passenger cars Hrafnkelsdóttir (2020). The present thesis does not test that operational model; it examines population-level accident occurrence using a separate data design.

Empirical work has compared the share of accidents occurring in high wind with the share of time that high wind was observed, an approach closely related to the observed/expected comparison used here Edwards (1994). Studies linking crashes to nearby weather stations likewise show that station observations can be informative while retaining measurement error Young and Liesman (2007). More broadly, reviews find that road-safety evidence for wind is less consistent than for precipitation Theofilatos and Yannis (2014).

2.2 Temperature, Daylight, and Time of Day

Temperature, daylight, and time of day add context that wind alone cannot provide. They affect road-surface conditions, visibility, travel patterns, and the composition of road users, but accident counts in these categories are not rates unless the corresponding amount of travel is known. Temperature effects can also be non-linear and season-specific: a Seoul time-series study found its strongest cold-weather association below freezing in winter, while an hourly case-crossover study in Shenzhen found a J-shaped temperature relationship Lee et al. (2014);

Zhan et al. (2020). Temperature is therefore compared with its local seasonal frequency rather than treated as having one linear effect.

Darkness presents the same denominator problem. A method developed using Norwegian, Swedish, and Dutch data compared daylight and darkness within clock time and season, avoiding the interpretation of raw night-time counts as risk Johansson et al. (2009). Here, raw hour, season, and daylight counts describe the sample. Matched daylight and adjusted severity models are used for numerical comparisons.

2.3 Traffic and Accident Rates

Weather can change the amount of traffic as well as the hazard. Research using automatic traffic recorders shows that extreme weather variables, including wind, can reduce or redistribute hourly traffic volume Sathiaraj et al. (2018). Accident counts during strong wind can remain low even if risk to each travelling vehicle increases. High-resolution crash studies show the value of hourly weather and matched-time analyses rather than daily or monthly averages Xing et al. (2019). The analyses distinguish three quantities: accident occurrence relative to weather frequency, rates based on estimated vehicle-kilometres, and changes in observed daily traffic. Studies with real-time traffic, weather, and road geometry show what a more complete analysis could achieve when such data are available Ahmed et al. (2012).

3 Data and Methods

3.1 Data

3.1.1 Source Data

Table 3.1 summarises the source datasets. Official documentation for each source is listed in the bibliography (Samgöngustofa, 2026; Veðurstofa Íslands, 2026; Vegagerðin, 2026; Hagstofa Íslands, 2026).

Table 3.1: Source datasets. The study period is 2007–2025.

Data	Provider	Contents	Coverage
Accident	Samgöngustofa	Accident events and road links	126,607 events and 126,610 road links, 2007–2025
Weather	Icelandic Meteorological Office (IMO)	Ten-minute weather observations and station metadata	232,459,562 observations from 338 stations; 438 station records, 2007–2025
Traffic and roads	Vegagerðin	Annual road traffic, daily counter reports, counter locations, and road geometry	33,757 section-years, 2000–2025; 1,033,659 counter channel-days, 2019–2024
Urban areas	Statistics Iceland	Urban-area polygons	97 polygons, 2020–2024 publication

The study includes 2025 accident, injury, vehicle, road-link, annual-traffic, and weather records under the same selection and quality criteria as earlier years. Daily counter data are available through 2024.

Table 3.2: Definitions used throughout the thesis

Term	Definition	Use in this thesis
f	Ten-minute mean wind speed, m/s	Main weather measure, matched within five minutes of the accident time.
fg	Wind gust reported in the same ten-minute observation, m/s	Additional weather measure; not a daily maximum.
t	Ten-minute air temperature, °C	Additional weather measure matched independently.
O/E	Observed accidents divided by accidents expected from local weather frequency	Main relative-occurrence result; traffic is not included.
ADU	Average vehicles per day over the year	Used to check and derive annual traffic quantities.
SDU	Average vehicles per day in June–September	Official summer traffic value.
VDU	Average vehicles per day in December–March	Official winter traffic value.
VH DU	Average vehicles per day in April–May and October–November	Spring/autumn daily traffic calculated as the residual from ADU, SDU and VDU.

3.1.2 Accident Data

The accident dataset contains 126,607 records from 2007–2025 Samgöngustofa (2026). One record lacks valid temporal or spatial information, leaving 126,606 valid records. Applying the rural classification described below yields 24,005 accidents, of which 6,414 involve injury. Of these, 6,259 have a qualifying primary weather match within 20 km and five minutes.

Accident severity is recorded in `meidsli`: code 1 denotes a fatal accident, code 2 a serious accident, code 3 a minor-injury accident, and code 4 an accident without injury. The main outcome is $\text{meidsli} \leq 3$. The serious-or-fatal comparison uses $\text{meidsli} \leq 2$. The source data also contain 17,591 rural damage-only accidents; these are outside the injury outcome.

Rural status is derived from accident coordinates and Statistics Iceland urban-locality polygons Hagstofa Íslands (2026). Points inside an urban polygon are classified as Urban, and valid points outside all polygons as Rural. A defined Vestmannaeyjar area is also classified as Urban. The same 2020–2024 polygon dataset is applied to the full 2007–2025 accident period. Changes in urban boundaries before 2020 are not represented.

3.1.3 Weather Data

The weather dataset contains 10-minute observations from Icelandic weather stations. The variables used here are ten-minute mean wind speed (`f`), reported wind gust (`fg`), and temperature (`t`). For each accident, the gust is taken from the same matched ten-minute observation as mean wind; it is not the maximum across the accident day. Mean wind speed remains the main weather measure. A wind observation is used when

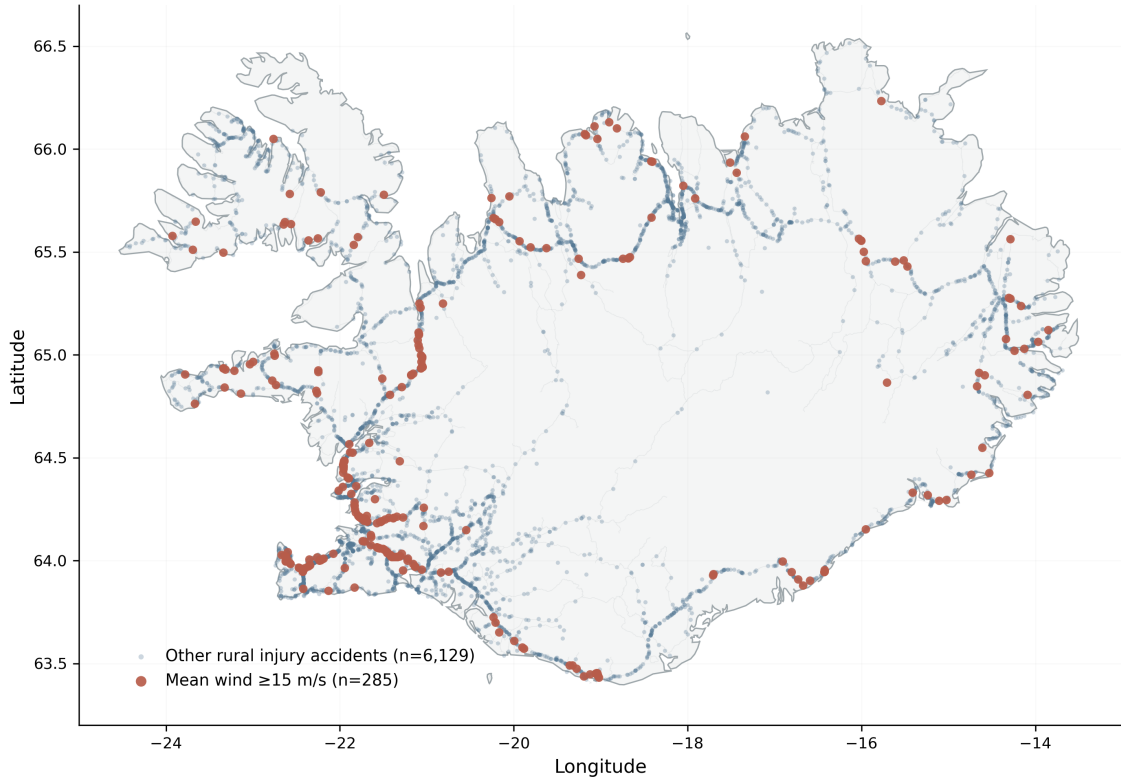


Figure 3.1: Locations of the 6,414 rural injury accidents. Accidents matched to mean wind of at least 15 m/s are highlighted. The map shows the spatial coverage of the study and is not a regional accident-rate map.

$$0 \leq f < 45, \quad 0 \leq fg < 65, \quad fg + 0.5 \geq f.$$

Rows missing either f or fg are excluded. Negative values, values at or above either upper limit, and internally inconsistent values where $fg = 0$ but $f > 0$ are excluded and reported in the annual audit. A value of $f = 0$ alone is kept. Only uninterrupted all-zero runs where $f = fg = 0$ for at least two hours are excluded as potentially frozen sensors. Missing temperature does not affect whether a wind observation is used.

Each accident is matched to the geographically nearest station with a quality-controlled wind observation at one of the two adjacent timestamps on the 10-minute observation grid. The observations have 10-minute resolution, but the selected observation is no more than five minutes from the recorded accident time. The maximum matching distance is 20 km. The matching dataset includes all 6,414 accidents and records whether each accident meets the weather-match criterion.

Temperature is matched independently within the same 20 km and five-minute limits. The nearest valid value between -30 and 30°C may therefore come from a different station when the wind station lacks temperature. In the final analysis dataset, however, all 6,259 wind and temperature matches use the same station and observation time.

The IMO supplied observations of time, station, temperature, mean wind, and gust. The same quality-controlled weather data are used for accident matching, background frequencies, matched control times, and traffic analyses. No temperature value is imputed.

Table 3.3 reports spatial and temporal match quality for wind and temperature.

Table 3.3: Share of accidents with weather information from a station with a valid measurement within 20 km. Distance is shown as median / P90; every retained observation is within five minutes of the accident time.

Variable	Accidents	Share	Stations	Distance (km)
Mean wind and gust	6,259	97.6%	279	5.1 / 12.3
Temperature	6,259	97.6%	279	5.1 / 12.3

Table 3.4: Wind-data scope and quality audit, 2007–2025. Quality-rule shares use all supplied observations. Every supplied station-year contains at least one wind measurement.

Category	Records	Share
All supplied station-time observations	232,459,562	100.00%
Missing f or fg	261,519	0.11%
Negative or upper-threshold wind	8,766	0.00%
Internally inconsistent f/fg	257,245	0.11%
Frozen all-zero runs (≥ 2 hours)	1,473,082	0.63%
Quality-controlled wind observations retained	230,458,950	99.14%

The wind-quality rules exclude 2,000,612 observations and retain 230,458,950 (99.14%). The largest excluded category is uninterrupted all-zero runs, which is reported separately because it reflects the rule for possible instrument stoppages rather than a missing observation. The audit reports each rule separately and uses all supplied weather observations from 2007–2025. For each accident’s station and season, the O/E expected count uses all quality-controlled background observations in that station–season group, not only observations recorded on accident days.

3.1.4 Traffic Data

Two traffic sources are used for different purposes.

Annual road-section data provide average annual daily traffic (ADU), summer daily traffic (SDU), winter daily traffic (VDU), section length, and annual vehicle-kilometres for 2000–2025 Vegagerðin (2026). The road-section comparison uses the years shared with the accident study, 2007–2025. Following the official definitions, SDU covers June–September and VDU covers December–March. These data can be linked to accidents with a registered road-section identifier, but exact matching is much narrower than the weather matching.

Table 3.5: Official traffic averages and the spring/autumn daily traffic value used in the annual-traffic analysis.

Period	Months	Daily traffic reference
ADU (<i>ársdagsumferð</i>)	Full calendar year	Official annual daily traffic
VDU (<i>vetrardagsumferð</i>)	December–March	Official winter daily traffic
SDU (<i>sumardagsumferð</i>)	June–September	Official summer daily traffic
VHDU (<i>vor- og haustdagsumferð</i>)	April–May and October–November	Residual daily average calculated from ADU, SDU, VDU, and the number of days in each period

Throughout the analyses, the four study seasons are aligned with these traffic periods: winter is December–March, spring is April–May, summer is June–September, and autumn is October–November. These are study-specific traffic-aligned seasons rather than conventional three-month meteorological seasons. VHDU is not calculated as $ADU - SDU - VDU$. Annual traffic implied by ADU is first converted to a calendar-year total; the day-weighted SDU and VDU totals are subtracted, and the remainder is divided by the number of spring and autumn days:

$$VHDU = \frac{D_{year}ADU - D_{VDU}VDU - D_{SDU}SDU}{D_{VHDU}}.$$

Here, each D is the number of calendar days in that period. VHDU is therefore a residual estimate rather than an additional official traffic measure.

Daily traffic counts extracted from Vegagerðin reports cover 2019–2024 and provide observed totals at selected counter sites. They support both the traffic-response comparison and secondary daily-counter accident-rate analyses, but do not provide hourly vehicle-kilometres at every accident location. More than one counter may occur on a road section, and counter positions are estimated from the registered road geometry. These data remain separate from the primary O/E calculation.

Of 68,946 annual road-section–year–period records, 60,595 have nearby usable wind data; separately, 738,424 of 774,274 daily counter-days have a matched daytime wind summary. The more restrictive accident samples used by the traffic models are reported in Chapter 4.

3.2 Data Preparation and Reproducibility

Table 3.6 summarises the sequence from source data to the reported results. Exact computational steps are documented separately.

Table 3.6: Data preparation and analysis workflow

Stage	Main data	Output	Role in thesis
Data preparation	Accident, weather, road and traffic data	Quality-controlled datasets	Defines the study data.
Accident selection	Accident data and urban boundaries	Rural injury accidents	Defines the study population.
Weather matching	Accidents, weather observations and stations	Accident weather, local frequencies and control times	Creates the weather comparisons.
Weather-frequency O/E	Accident weather and local frequencies	Observed and expected counts	Primary analysis.
Matched-time analysis	Accident and control times	Matched odds ratios	Time-matched supporting comparison.
Traffic analyses	Accident weather, traffic and road lengths	Traffic-linked rates	Supporting analyses in smaller samples.

3.3 Analysis

3.3.1 Weather-Frequency Comparison

Accident-time mean wind speed f is the main weather measure and is divided into 5 m/s intervals. It is the ten-minute mean wind observation matched within five minutes of the accident time, not an instantaneous measurement. Wind gust f_g from the same observation and independently matched temperature are additional weather measures. For each measure, an accident is compared with the distribution at its matched station in the same study season, pooled across all study years. Expected counts are based on how often each condition occurred locally rather than on a national weather distribution. For example, if 5% of the relevant station–season observations fall in one wind interval, approximately 5% of that group’s accidents would be expected there under weather-frequency-proportional occurrence.

Let g identify a weather station and study season. Let A_{jg} be the number of accidents in weather interval j and group g , and let $A_g = \sum_j A_{jg}$. From all quality-controlled 10-minute observations, let T_{jg} be the number in interval j and T_g the total in group g . The local background frequency is

$$p_{jg} = \frac{T_{jg}}{T_g}.$$

If accident timing followed local wind frequency, the expected number in the interval would be $A_g p_{jg}$. The pooled observed/expected ratio is

$$R_j = \frac{\sum_g A_{jg}}{\sum_g A_g p_{jg}}.$$

A value of one means that the accident share equals the local time share of the weather interval. A value above one means that accidents are over-represented in that interval. Standardisation

within station and season prevents locations or seasons with different weather distributions from dominating the result.

The calculation is made for all injury accidents ($meiðsli \leq 3$) and for the serious-or-fatal subset ($meiðsli \leq 2$). The groups therefore overlap by design: the first shows total injury-accident occurrence and the second shows whether the same pattern is visible among the more severe accidents. The O/E values are descriptive and are therefore shown without confidence intervals. Wind gust and temperature are handled in the same way as mean wind. Temperature is matched independently, and its intervals are below -6 , -6 to -3 , -3 to 0 , 0 to 3 , 3 to 6 , 6 to 9 , 9 to 12 , and at least 12°C . The O/E upper interval is combined at 12°C to keep the seasonal panels readable. The matched-time and annual-traffic models retain separate 12 – 15 and at-least- 15°C intervals. Estimates across those different upper intervals are not treated as direct numerical comparisons.

3.3.2 Matched-Time Comparison

This is a time-stratified case-crossover design: each accident forms one matched set. The accident time is the case time, and the control times are the other occurrences of the same weekday and clock time in the same calendar month and year. For example, a Tuesday accident at 14:20 is compared with the other Tuesdays at 14:20 in that month. Mean-wind and gust controls use the accident's matched wind station; temperature controls use its independently matched temperature station. Control weather must satisfy the same five-minute observation-time rule, and both station matches are restricted to 20 km. Conditional logistic regression compares weather at the accident time with weather at its matched control times. This time-stratified referent strategy was selected to avoid overlap bias and to control calendar time without selecting controls from outside the accident month Janes et al. (2005).

The model asks whether weather at an accident time differs from weather at comparable non-accident times. Matching controls exactly for station, year, month, weekday, and clock time, but not for unusual traffic, road conditions, or other factors that change on a particular day. Continuous effects are reported per 5 m/s or 5°C together with categorical estimates. The mean-wind O/E remains the primary result.

Mean wind and temperature are also entered together as categorical variables to assess whether the wind estimate changes after including temperature at the same case and control times. Daylight is calculated at the accident location for the accident date and the same matched dates. Because month, weekday, and clock time are held fixed, only strata that change between darkness, civil twilight, and daylight contribute information.

Seasonal differences in the mean-wind association are tested in a separate matched-time model. Mean wind is grouped as 0–10, 10–15, and at least 15 m/s for stability. The model includes interactions between wind interval and season, and a likelihood-ratio test compares it with a model having one common wind association across seasons. The omnibus test is used instead of judging seasonal differences from separate confidence intervals.

A separate logistic model describes serious-or-fatal rather than minor injury among accidents with complete wind and temperature matches. It includes broad categories of mean wind, temperature, daylight, time of day, and season. The result concerns injury severity among recorded accidents. It is not a model of whether an accident occurs and does not replace the traffic analyses.

3.3.3 Use of Traffic Data

The primary O/E denominator is the local frequency of wind conditions rather than traffic volume. Complete traffic data are unavailable at the required spatial and temporal resolution. O/E therefore does not account for fewer journeys on windy days. If traffic falls in strong wind while other factors remain unchanged, the accident-rate contrast per vehicle-kilometre would be larger than the O/E contrast. The traffic analyses examine this possibility. The descriptive estimated-rate calculation uses annual traffic on every eligible road section, including sections without an accident. The main conditional model has a narrower purpose and sample: it includes 4,933 accidents in road-section–year–traffic-period strata that contain an accident and have a valid weather and traffic link. Both calculations use SDU and VDU where they are published, and the residual VHDU estimate for April–May and October–November.

Adding these traffic data to the primary O/E calculation would replace the broad study population with a smaller linked sample and combine seasonal traffic averages with ten-minute weather. O/E is therefore interpreted as relative accident occurrence, not accident rate per vehicle.

3.3.4 Road-Section Traffic Comparison

Traffic is included when accidents can be linked by road section, year, official traffic period, and wind interval k, j , estimated vehicle-kilometres are

$$V_{kj} = q_k L_k D_k p_{kj},$$

where q_k is SDU, VDU, or derived VHDU vehicles per day, L_k is section length, D_k is the number of calendar days, and p_{kj} is the local share of quality-controlled 10-minute observations in the wind interval. The accident rate is

$$\text{Rate}_j = \frac{A_j}{\sum_k V_{kj}} \times 100,000,000.$$

This method accounts for road length and for spatial and seasonal differences in published traffic. It still assumes that traffic within a traffic period is distributed across wind intervals in proportion to time. It does not observe traffic at the accident minute. The station nearest the road-section midpoint with qualifying data supplies p_{kj} . Each accident is matched to that road-section station rather than the accident-nearest station used in the primary O/E analysis. An accident is included only when the road-section station is within 20 km and has a valid observation within five minutes. The accident count and its estimated denominator therefore use the same weather station.

The main traffic result is a conditional Poisson model. Each road section, year, and traffic period has its own intercept. For stratum k and wind interval j , the model is

$$\log E(A_{kj}) = \alpha_k + \beta_j + \log V_{kj},$$

where A_{kj} is the accident count and $\log V_{kj}$ is the offset. The exponentiated coefficient

$\exp(\beta_j)$ is the estimated rate ratio relative to 0–5 m/s within otherwise identical road-section–year–period strata.

The model is fitted separately for all injury accidents and for serious or fatal accidents. A seasonal extension uses year-specific station frequencies for winter, spring, summer, and autumn. Because individual 5 m/s upper bins are sparse within seasons, that extension uses 0–10, 10–15, and at least 15 m/s. Winter and summer use published VDU and SDU, respectively; spring and autumn use the same derived VH DU daily volume but retain separate weather frequencies and accident counts.

The descriptive rate and conditional model include different road records. The descriptive rate sums estimated vehicle-kilometres over every eligible road section. The conditional model includes only road-section–year–period strata containing at least one accident, because all-zero strata provide no within-stratum information: after conditioning on a stratum’s total accident count, a total of zero contributes no likelihood for comparing its wind intervals. Those roads remain in the separate descriptive rate that sums all eligible exposure. Wind intervals with positive estimated vehicle-kilometres are kept within those groups. The two traffic results answer different questions and are not expected to reproduce the same rate ratios.

3.3.5 Daily Traffic Comparison

Daily traffic is analysed separately using daily weather summaries. Directional channels are first summed to one 24-hour count for each physical counter and date. The unit is then one fixed counter on one date; separate counter sites on the same road section are not summed. For each counter, year, month, and weekday, expected traffic is the arithmetic mean of the available daily counts in that same group. The traffic index is

$$I_{cd} = 100 \times \frac{Q_{cd}}{\text{mean}(Q_{c, \text{same year, month, weekday}})}.$$

An index below 100 indicates fewer counted vehicles than the mean for that local calendar context. Results are aggregated across counter-days and describe the participating sites rather than national traffic. The 24-hour count is not converted into hourly traffic. Mean wind is calculated from quality-controlled observations between 10:00 and 21:59, so this is a daily association rather than an estimate of traffic in that window. The daily-counter dataset contains 5,432 counter-days with a reported count of zero. They are kept because the reports do not distinguish a genuine zero or road closure from counter downtime; the appendix reports a check excluding these records. For each located counter, the nearest station within 20 km having usable observations that day is selected.

Sustained wind is described by the number of valid ten-minute observations in each counter-day with mean wind of at least 15 m/s; six observations equal one hour. Days must contain at least 108 valid observations. Traffic is compared across 0, more than 0–2, 2–6, and at least 6 hours of strong wind using the same counter–calendar standardisation.

The allocated daily-counter rate model uses a stricter accident linkage. Each 2019–2024 accident requires a counter on the exact registered road section and year within 20 km, positive traffic on its date, and qualifying accident-time mean wind. For counter (c), date (d), and

wind interval (j), the observed 24-hour count is allocated according to the counter station's valid ten-minute observations:

$$\hat{Q}_{cdj} = Q_{cd} \frac{N_{cdj}}{N_{cd}}.$$

Accidents are classified by accident-time f from the same station used for the counter-day denominator. That station must be within 20 km of both the accident and counter, and the accident observation must be within five minutes of the recorded time. Estimated traffic and accidents are aggregated by counter, year, and wind interval, and a conditional Poisson model gives each counter-year its own intercept. The allocation assumes uniform traffic across valid observation times; it is an estimated within-day traffic split, not observed hourly traffic. The main allocation uses all sufficiently complete 24-hour weather records.

A second daily-counter calculation uses the counter-section construction. For each road section and day, the observed daily vehicle count is multiplied by the road length assigned to that counter. This daily vehicle-kilometre total is allocated among weather intervals using the actual 10-minute observations at the assigned station from 07:00 to midnight on that same date. Accidents outside this time window are omitted. At least 92 of the expected 102 observations are required. For day i , counter-section j , and interval b ,

$$\hat{V}_{ijb} = Q_{ij} L_j \frac{N_{ijb}}{N_{ij}},$$

where Q_{ij} is the observed 24-hour count, L_j is the assigned section length, N_{ijb} is the number of valid same-day observations in interval b , and N_{ij} is the number of valid observations that day. The accident rate is the observed accidents divided by summed $\hat{V}_{ijb}$. This compares accidents per estimated vehicle-kilometre, but the weather-bin allocation is estimated: hourly traffic is not observed. Unlike the annual method, observed traffic on a windy date contributes its own daily total, so lower travel on that date is reflected in the denominator.

4 Results

Results follow the three research questions: weather-frequency O/E, matched-time comparisons, and traffic analyses in smaller linked samples.

4.1 Study Sample and Coverage

Table 4.1: Accident selection and analysis sample sizes.

Stage	Accidents	Purpose
Source records	126,607	Accident database, 2007–2025.
Rural injury accidents	6,414	Study population.
Primary O/E	6,259	Main weather analysis.
Matched-time	6,257	Time-matched comparison.
Annual traffic	4,933	Broader traffic analysis.
Daily-counter	762	Observed daily traffic totals.
Same-day vehicle-kilometres	613	Stricter counter-section analysis.

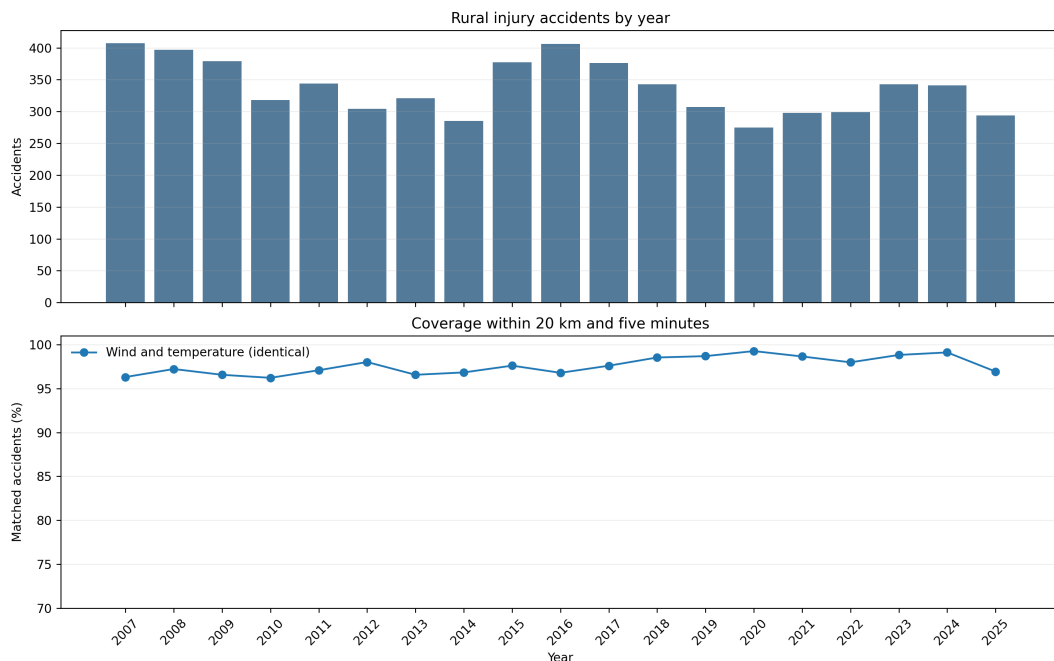


Figure 4.1: Annual rural injury-accident counts and the percentage matched to wind or temperature within 20 km and five minutes. The wind and temperature match counts are identical in every year; no missing temperature value is imputed.

The 20 km threshold includes most accidents while limiting the spatial distance to the weather proxy. Increasing the threshold to 30 km adds 207 accidents, but the greater distance may increase weather-exposure misclassification in complex terrain. The road-section analysis is much narrower than the main analysis and is less geographically representative.

Wind and temperature are matched independently, but in the final analysis dataset both procedures retain the same 6,259 accidents and select the same station and observation time for each of them. Within 2019–2024, 762 of 1,863 accidents enter the allocated daily-rate model after requiring the exact road section, a valid observed counter-day, the same weather station for the accident and denominator, and all 20 km distance limits. The broader daily-counter selection step includes 767 accidents and is documented in the appendix.

4.2 Accident Frequency by Weather

Each of the following figures shows one weather measure for the full year and the four seasons. Blue bars show all injury accidents and orange bars show their serious-or-fatal subset. Counts are printed above the bars. Seasonal estimates in the upper weather intervals are descriptive because several contain few accidents.

4.2.1 Mean Wind

Ratios are close to one across low and moderate intervals and increase from 15 m/s. At 20 m/s or above, 68 injury accidents were observed compared with 31.6 expected (O/E 2.15). The serious-or-fatal subset contained 12 accidents compared with 6.4 expected (O/E 1.87). These are descriptive ratios; the smaller serious-or-fatal count does not support a precise statement about whether severity changes with wind.

The 5 m/s grouping is used throughout because it gives a direct, readable description of the upper wind intervals without implying that the boundaries are physical thresholds.

4.2.2 Wind Gust

Accident-time wind gust is an additional weather measure rather than a daily maximum. At 30 m/s or above, 61 injury accidents were observed compared with 18.2 expected (O/E 3.34). The serious-or-fatal subset contained nine accidents compared with 3.6 expected (O/E 2.47). Gust and mean wind are related measures and should not be treated as independent findings.

4.2.3 Temperature

Temperature is interpreted cautiously. Valid temperature was available for 6,259 accidents (97.58%). Coverage by year is shown in Figure 4.1, and no temperature value is imputed. After adjustment by temperature station and season, the estimates were non-monotonic. O/E was below one at 3–6°C for all injury accidents (808 observed and 1,137.0 expected; O/E 0.71), but above one at -3–0°C. At 12°C or above, 789 injury accidents were observed compared with 493.8 expected (O/E 1.60). This pattern may reflect travel behaviour or

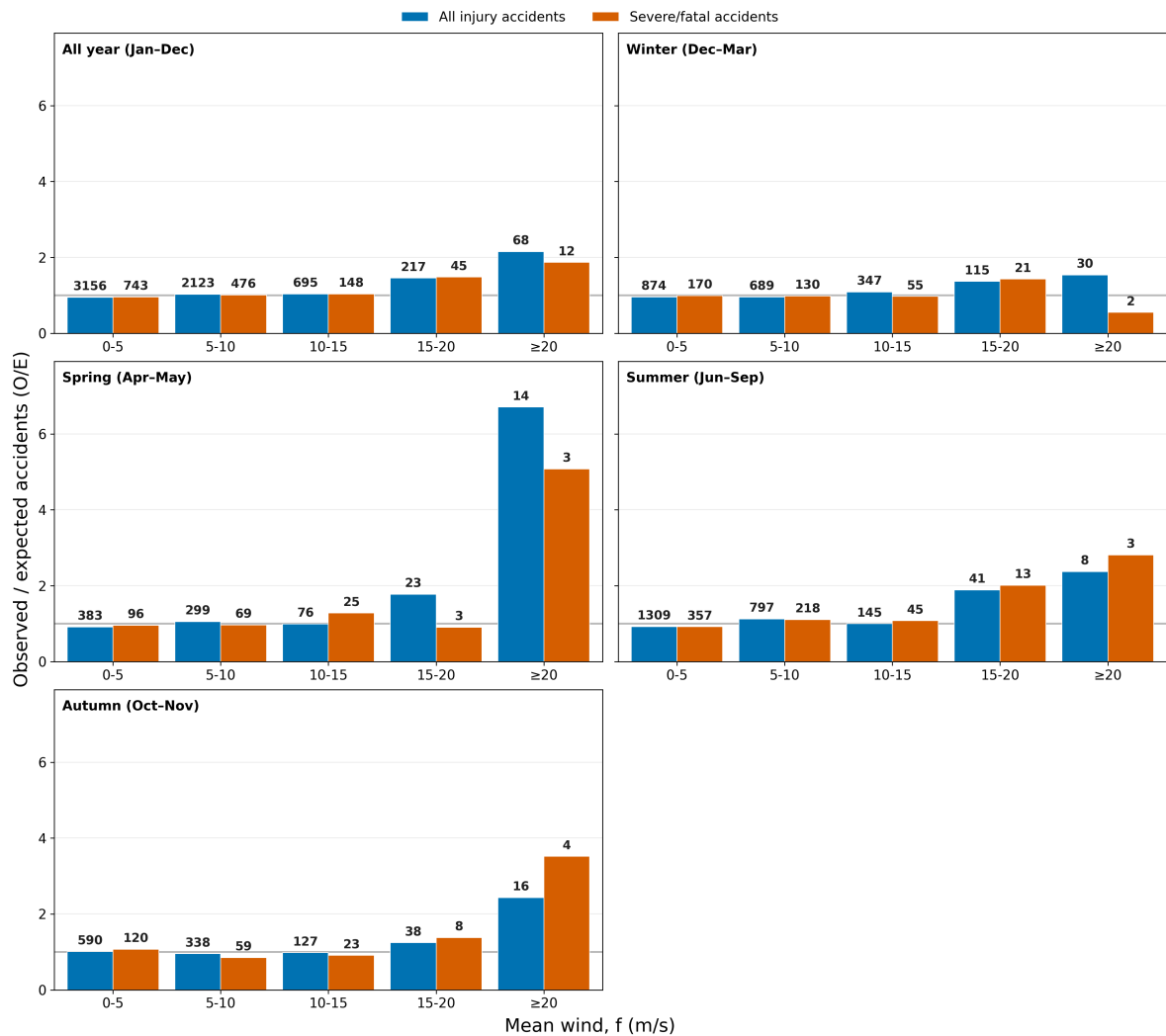


Figure 4.2: All-year and seasonal mean-wind O/E, 2007–2025. Blue bars include all accidents with injury ($mei\ddot{s}li \leq 3$); orange bars show the serious-or-fatal subset ($mei\ddot{s}li \leq 2$). Expected counts use local weather frequency; numbers above bars are observed accidents.

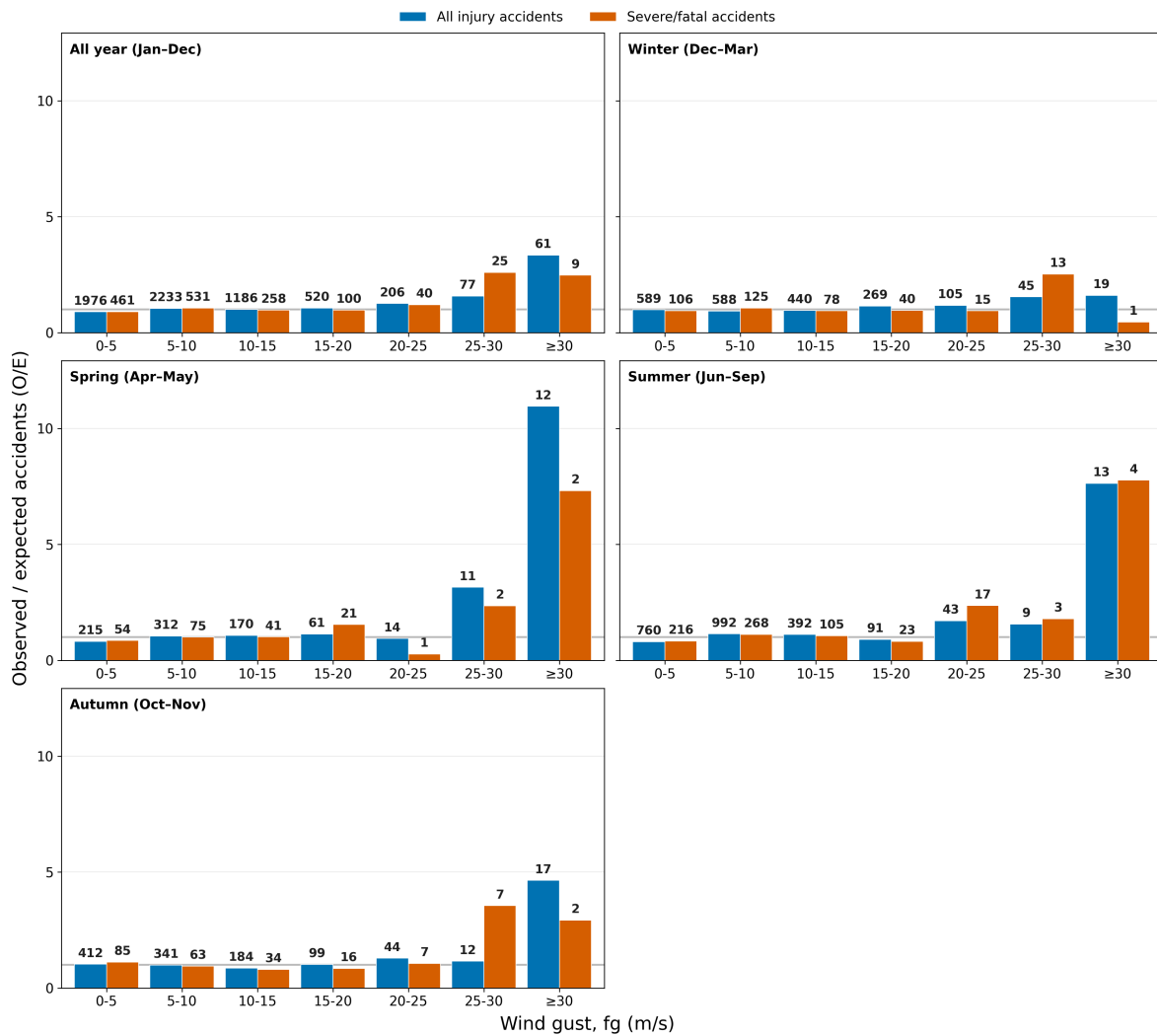


Figure 4.3: All-year and seasonal wind-gust O/E. Gust is taken from the ten-minute observation matched to the accident time; it is not a daily maximum. Orange bars are a subset of the blue bars; numbers above bars are observed accidents.

weather and road conditions associated with temperature. It is not interpreted as a causal temperature effect. For the serious-or-fatal group, all-year O/E is 0.69 at 3–6°C and 1.71 at 12°C or above (176 and 226 accidents, respectively).

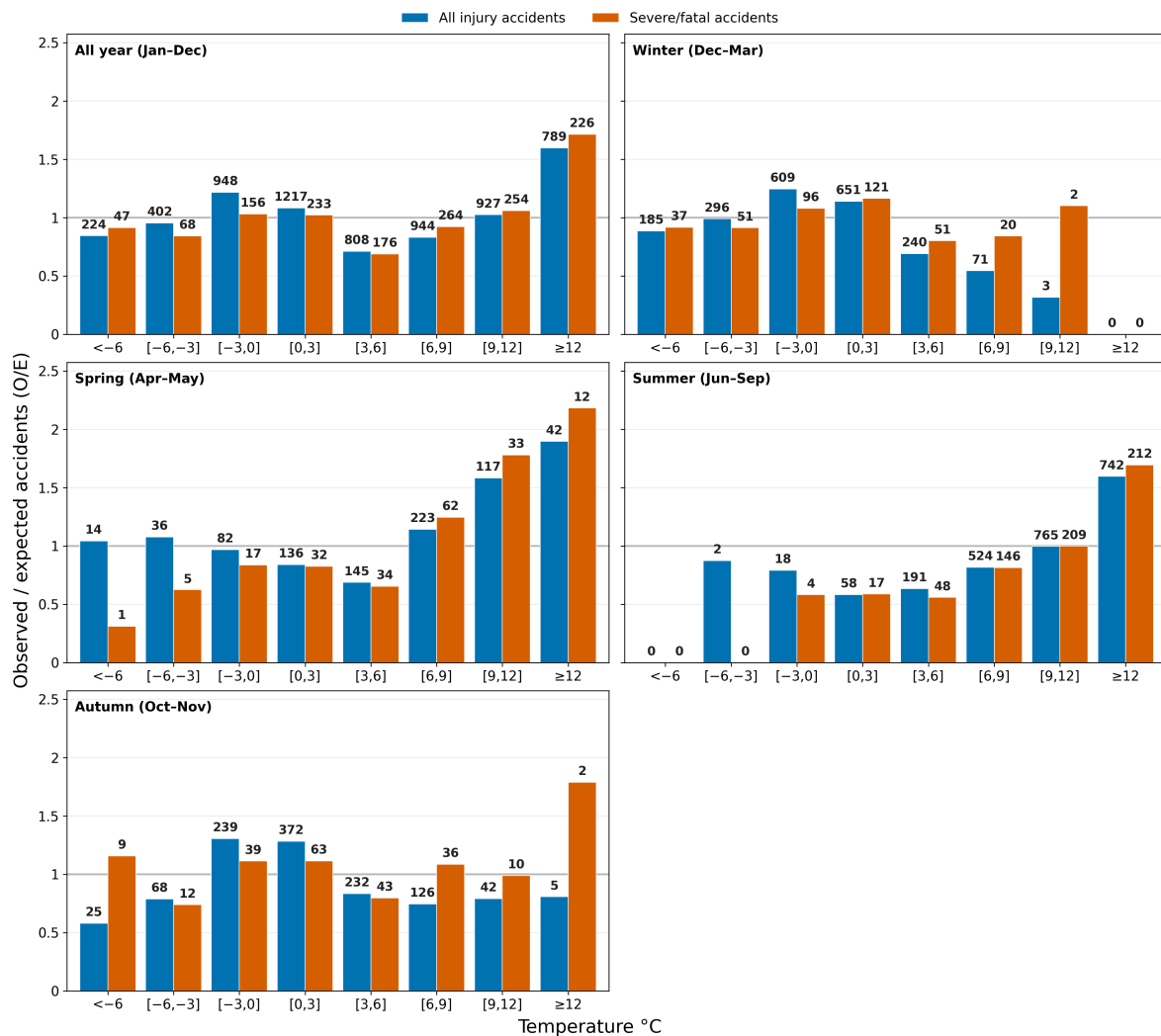


Figure 4.4: All-year and seasonal temperature O/E, standardised by matched temperature station and study season. Serious-or-fatal accidents are a subset of all injury accidents; numbers above bars are observed accidents. The non-monotonic pattern is interpreted cautiously.

Adding year to the station–season standardisation changes the selected estimates by no more than 0.02: for example, mean-wind O/E at 20 m/s or above changes from 2.15 to 2.17. The pre-specified station–season result is retained.

4.3 Matched-Time Results

The case-crossover analysis includes 6,257 mean-wind and gust strata with 21,139 control times for each measure, and 6,257 temperature strata with 21,144 control times. A 5 m/s increase in mean wind is associated with a matched odds ratio of 1.09 (95% CI 1.05–1.13). For categorical mean wind, the clearest increase occurs at 15 m/s or above: the odds ratio

relative to 0–5 m/s is 1.61 (95% CI 1.38–1.87). This supports the high-wind pattern after exact control for station, year, month, weekday, and clock time.

For gust, the matched odds ratio per 5 m/s is 1.06 (95% CI 1.04–1.09). Relative to 0–10 m/s, it is 1.36 (1.15–1.60) at 20–25 m/s, 1.72 (1.30–2.28) at 25–30 m/s, and 2.74 (1.96–3.85) at 30 m/s or above. The matched-time result agrees with the upper-gust O/E pattern.

The temperature estimates are not monotonic. Relative to 0–3°C, the matched odds ratio is 1.13 (95% CI 1.02–1.25) at -3–0°C, 0.59 (0.53–0.65) at 3–6°C, and 0.99 (0.80–1.23) at 15°C or above. This last interval is narrower than the O/E at-least-12°C interval. The matched-time result reproduces the dip at moderate temperatures but not the warm-tail increase in the O/E analysis. The methods therefore give different results at warm temperatures, which calls for caution.

When mean wind and temperature are included together, the matched odds ratio for mean wind of at least 15 versus 0–5 m/s is 1.68 (95% CI 1.44–1.96). This is close to the single-variable estimate of 1.61, and the non-monotonic temperature pattern remains.

4.4 Daylight, Season, and Severity

The matched-date daylight comparison holds location, month, weekday, and clock time fixed. Relative to daylight, the odds ratio is 1.14 (95% CI 0.88–1.49) for darkness and 1.10 (0.92–1.31) for civil twilight. Only 985 of the 6,414 strata change daylight class within their matched dates. The result provides no clear evidence of a daylight association and is much less informative than the wind analysis; it does not use vehicle counts.

Among the 6,259 accidents in the adjusted severity model, 1,424 are serious or fatal. Mean wind of at least 15 m/s has an odds ratio of 0.93 (95% CI 0.69–1.26) relative to 0–10 m/s. Darkness has an odds ratio of 0.89 (0.71–1.12) relative to daylight. Thus the data show a clear strong-wind pattern for accident occurrence, but not for severity conditional on a recorded injury accident. The complete adjusted estimates are shown in Table A.1 in the appendix.

The matched-time wind-by-season comparison uses the same 6,257 mean-wind strata as the case-crossover analysis, with coarse intervals for stability. The omnibus interaction test gives $\chi^2(6) = 14.30$, $p = 0.026$. The estimated odds ratios for at least 15 versus 0–10 m/s are 1.44 in winter, 2.02 in spring, 2.53 in summer, and 1.37 in autumn. The omnibus test provides evidence that at least some seasonal wind associations differ, but it does not identify a particular pair of seasons as different. Season-specific estimates are therefore not ranked.

4.5 Traffic Results

The annual analysis has broader coverage, whereas the daily-counter analysis uses observed daily totals at fewer locations. Both require traffic to be allocated among wind intervals; neither observes traffic at the accident minute.

Table 4.2: Traffic inputs for the supporting vehicle-kilometre analyses.

Analysis	Years (accidents)	Traffic basis	Main limitation
Annual road sections	2007–2025 (4,933)	VDU, SDU and derived VHDU	Traffic is allocated by wind frequency within each period.
Daily counter sections	2019–2024 (613)	Observed daily totals	Within-day traffic is estimated; locations and upper bins are sparse.

4.5.1 Broader Annual-Traffic Comparison

Table 4.3: Estimated rural injury-accident rate by mean wind speed, all traffic periods

Mean wind	Accidents	Estimated accidents per 100 million vehicle-km
0–5	2,424	8.5
5–10	1,684	13.0
10–15	578	16.6
15–20	184	27.7
20–25	45	40.5
≥25	18	58.6

The descriptive rate rises across the wind intervals. After comparison within road section, year, and traffic period, the rate ratio for 20–25 versus 0–5 m/s is 2.27. The descriptive and conditional estimates differ because they account for road-section and period differences in different ways.

The conditional Poisson model includes 4,933 accidents in 3,984 road-section–year–traffic-period groups that contain accidents. Relative to 0–5 m/s, the estimated within-stratum rate ratio is 1.63 (95% CI 1.39–1.90) at 15–20 m/s, 2.27 (1.68–3.07) at 20–25 m/s, and 4.95 (3.07–7.98) at 25 m/s or above. The uppermost estimate contains 18 accidents and is less precise. Traffic is allocated within each period according to local wind frequency.

The serious-or-fatal model includes 1,055 accidents. Relative to 0–5 m/s, its rate ratio is 1.92 (95% CI 1.40–2.64) at 15–20 m/s and 2.23 (1.14–4.37) at 20–25 m/s. The at-least-25 m/s estimate is 4.91 (1.76–13.64), but contains only four accidents and is correspondingly imprecise.

Using wider seasonal intervals, the estimated rate ratio for at least 15 versus 0–10 m/s is above one in all four seasons. These values are supporting checks only: spring and autumn use derived VHDU, and the upper seasonal cells are too small for a precise ranking.

4.5.2 Daily-Counter Checks

Observed daily traffic falls as strong wind persists. Relative to the calendar expectation at the same counter, traffic is 86.8% on days with at least six hours at $f \geq 15$ m/s. Lower traffic during strong wind cannot account for the elevated O/E; if the selected counters reflect the wider road network, it would tend to make a per-vehicle contrast larger. The counters are geographically selective, however, and do not provide complete hourly traffic at accident times.

The allocated daily-counter model retains 762 accidents. Its rate ratio for at least 15 versus 0–10 m/s is 3.62 (95% CI 2.65–4.94), based on 47 accidents in the upper interval. A stricter counter-section calculation includes road length and actual same-day weather for 613 daytime accidents. Its upper mean-wind interval contains only ten accidents. Both calculations start from observed daily totals but estimate their within-day distribution, and both are limited to selected counter locations. They are therefore supporting checks, not the primary result.

Appendix Table A.5 addresses the two most important data questions. Restricting the rate model to the official VDU and SDU periods reduces the upper estimates, but they remain above one. Excluding daily counter records reported as zero changes the 20–25 m/s traffic estimate from 68.9% to 69.3%, so those records do not explain the observed decline.

Table 4.4: Main result and supporting comparisons.

Analysis	Estimate	Role
Weather-frequency O/E	≥ 20 m/s: O/E 2.15	Primary result; local wind frequency, no traffic.
Matched time	≥ 15 vs 0–5 m/s: OR 1.61 (1.38–1.87)	Same station and calendar time.
Annual traffic	20–25 vs 0–5 m/s: RR 2.27 (1.68–3.07)	Broader traffic sample; traffic allocated within periods.
Same-day traffic	≥ 20 vs 0–5 m/s: 88.5 vs 6.8 per 100 million VKT	Observed daily totals; 613 accidents, 10 in the upper interval.

The estimates measure different quantities and should not be compared by magnitude. Their direction is consistent, but hourly traffic is unavailable for the hour-of-day and daylight analyses.

5 Discussion

5.1 Interpretation

The primary O/E analysis shows that strong mean wind is over-represented at the times of rural injury accidents relative to its local frequency. The matched-time analysis supports this pattern after controlling for station and recurring calendar time. The traffic results point in the same direction but come from smaller samples and estimated exposure. The evidence therefore supports an association, not a causal effect or a national per-vehicle risk.

Observed daily traffic is lower when strong wind persists, so reduced travel does not explain the elevated O/E. However, neither traffic source observes traffic at the accident minute: the annual analysis allocates seasonal traffic by wind frequency, and the daily analyses allocate observed totals within the day. These results support rather than replace the primary comparison.

Gust also has an upper-tail pattern, although it is closely related to mean wind. Temperature is non-monotonic and differs at the warm end between the O/E and matched-time analyses. The daylight comparison gives no clear association, and the severity model gives no clear evidence that strong wind increases severity among recorded injury accidents. Sparse seasonal upper intervals do not support ranking seasons or outcome groups.

The main result agrees with earlier studies comparing accident shares with the time spent in adverse weather Edwards (1994) and with the engineering evidence that crosswind can impair vehicle stability Baker (1986, 1994). This study adds a longer Icelandic series, local station-and-season standardisation, and matched-time and traffic checks; it does not establish a new physical wind threshold. The non-monotonic temperature estimates are consistent with previous evidence that temperature associations can vary by range and season Lee et al. (2014); Zhan et al. (2020). The decrease in daily traffic during sustained wind is consistent with recorder-based evidence that adverse weather changes travel demand Sathiaraj et al. (2018).

5.2 Strengths

The study combines a long accident series with ten-minute weather observations, local weather-frequency standardisation, and explicit reporting of match coverage and observed counts. The matched-time and traffic analyses address different weaknesses of the primary comparison. Documented transformations, exclusions, and automated checks make the reported results reproducible.

5.3 Limitations

- The main result is not traffic-adjusted. It should not be interpreted as accident risk per vehicle-kilometre.
- The case-crossover analysis controls recurring calendar time but not unusual traffic, road conditions, visibility, or closures on a particular day. Its ≥ 15 m/s category summarises the upper-wind pattern rather than a physical threshold.
- A weather station up to 20 km away is a proxy for conditions at the accident location. Terrain can produce important differences over shorter distances.
- The annual traffic analysis allocates seasonal traffic across wind intervals and does not observe traffic at the accident minute.
- Daily counters cover selected sites in 2019–2024 and provide no hourly traffic. Their totals are allocated within the day, and an assigned counter need not have been passed by the accident vehicle.
- Estimated rates are restricted to rural injury accidents on linked road sections. They should not be compared directly with broad national or international accident rates unless the outcome and vehicle-kilometre traffic quantity are defined in the same way.
- The highest wind intervals contain relatively few accidents. Several intervals and comparison results are shown, so individual estimates in the upper tail should be interpreted with caution.
- Fixed 2020–2024 urban boundaries are applied to the entire 2007–2025 accident period.
- The analysis is observational. Unmeasured road condition, vehicle type, speed, visibility, and driver behaviour can contribute to the pattern.

5.4 Future Work

The most valuable addition would be verified hourly traffic with counter coordinates, road-section identifiers, direction information, and documented downtime. Linked to the same weather stations, these data could distinguish vehicle classes and wind direction relative to the road. Daily totals cannot provide that information.

6 Conclusion

Of 6,414 rural injury accidents during 2007–2025, 6,259 were matched to valid weather observations within 20 km and five minutes. At mean wind speeds of at least 20 m/s, 68 injury accidents occurred compared with 31.6 expected from local weather frequency (O/E 2.15). The serious-or-fatal subset had O/E 1.87, but its smaller count limits comparison between severity groups.

The matched-time analysis confirms the upper-wind pattern: the odds ratio for at least 15 m/s versus 0–5 m/s was 1.61. Gust gives a similar upper-tail pattern, whereas the non-monotonic temperature estimates differ between the O/E and matched-time analyses at warmer temperatures. The temperature finding therefore remains uncertain.

Annual and daily traffic analyses support the high-wind pattern, and observed daily traffic is lower during sustained strong wind. These analyses cover narrower, selected samples and estimate traffic within wind intervals from seasonal or daily totals; they do not provide complete hourly exposure at accident times.

Strong mean wind is therefore associated with higher relative occurrence of rural injury accidents in Iceland. This is not a causal estimate or a complete national measure of accident risk per vehicle. Such a measure would require linked hourly traffic data with wider geographic coverage.

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A Supporting Checks

This appendix contains the checks most relevant to interpreting the main results. Additional checks are documented separately.

A.1 Adjusted Severity Model

Table A.1 shows the adjusted comparison of serious-or-fatal and minor-injury accidents. It describes severity among recorded injury accidents; it does not estimate accident occurrence or risk per vehicle.

Table A.1: Adjusted odds ratios for a serious-or-fatal outcome among recorded injury accidents. All listed variables are included in the same model.

Variable	Comparison	Adjusted OR (95% CI)
Mean wind	10-15 m/s vs 0-10 m/s	0.99 (0.82–1.21)
Mean wind	≥ 15 m/s vs 0-10 m/s	0.93 (0.69–1.26)
Temperature	< -6 °C vs 0-3 °C	1.13 (0.79–1.62)
Temperature	-6 – -3 °C vs 0-3 °C	0.85 (0.63–1.15)
Temperature	-3 – 0 °C vs 0-3 °C	0.83 (0.66–1.04)
Temperature	3 – 6 °C vs 0-3 °C	1.11 (0.88–1.39)
Temperature	6 – 9 °C vs 0-3 °C	1.41 (1.10–1.79)
Temperature	9 – 12 °C vs 0-3 °C	1.28 (0.98–1.68)
Temperature	12 – 15 °C vs 0-3 °C	1.33 (0.98–1.81)
Temperature	≥ 15 °C vs 0-3 °C	1.21 (0.83–1.76)
Daylight	Darkness vs Daylight	0.89 (0.71–1.12)
Daylight	Civil twilight vs Daylight	1.09 (0.87–1.36)
Time of day	00-06 vs 10-16	0.86 (0.64–1.16)
Time of day	06-10 vs 10-16	0.65 (0.52–0.82)
Time of day	16-20 vs 10-16	0.98 (0.84–1.14)
Time of day	20-24 vs 10-16	0.77 (0.61–0.98)
Season	Winter vs Summer	0.82 (0.64–1.05)
Season	Spring vs Summer	0.96 (0.78–1.18)
Season	Autumn vs Summer	0.83 (0.65–1.05)

A.2 Daily-Counter Sample

The following tables show the narrowing of the 2019–2024 accident sample and compare retained accidents with those excluded by the daily-counter linkage. The 767 records at the last broad selection step include five records that do not enter the final allocated-rate model; that model retains 762 accidents.

Table A.2: Selection of accidents for the observed daily-counter analysis

Selection step	Accidents	Share of 2019–2024 sample
Rural injury accidents in 2019–2024	1,863	100.0%
Exact road-section/year counter candidate	864	46.4%
Assigned counter within 20 km	863	46.3%
Positive count and valid full-day wind	767	41.2%

Table A.3: Characteristics of accidents retained and excluded by the daily-counter linkage

Characteristic	All	No link	Linked, excluded	Retained
Accidents	1,863	999	102	762
Share of all accidents	100.0%	53.6%	5.5%	40.9%
Distinct road sections	563	374	56	189
Serious or fatal	25.6%	29.4%	18.6%	21.5%
One vehicle	67.7%	75.2%	54.9%	59.7%
Winter	32.6%	28.0%	40.2%	37.5%
Spring	13.0%	11.7%	8.8%	15.4%
Summer	36.0%	41.8%	33.3%	28.6%
Autumn	18.4%	18.4%	17.6%	18.5%

The geographic selectivity of this traffic-linked sample is documented in Figure A.1.

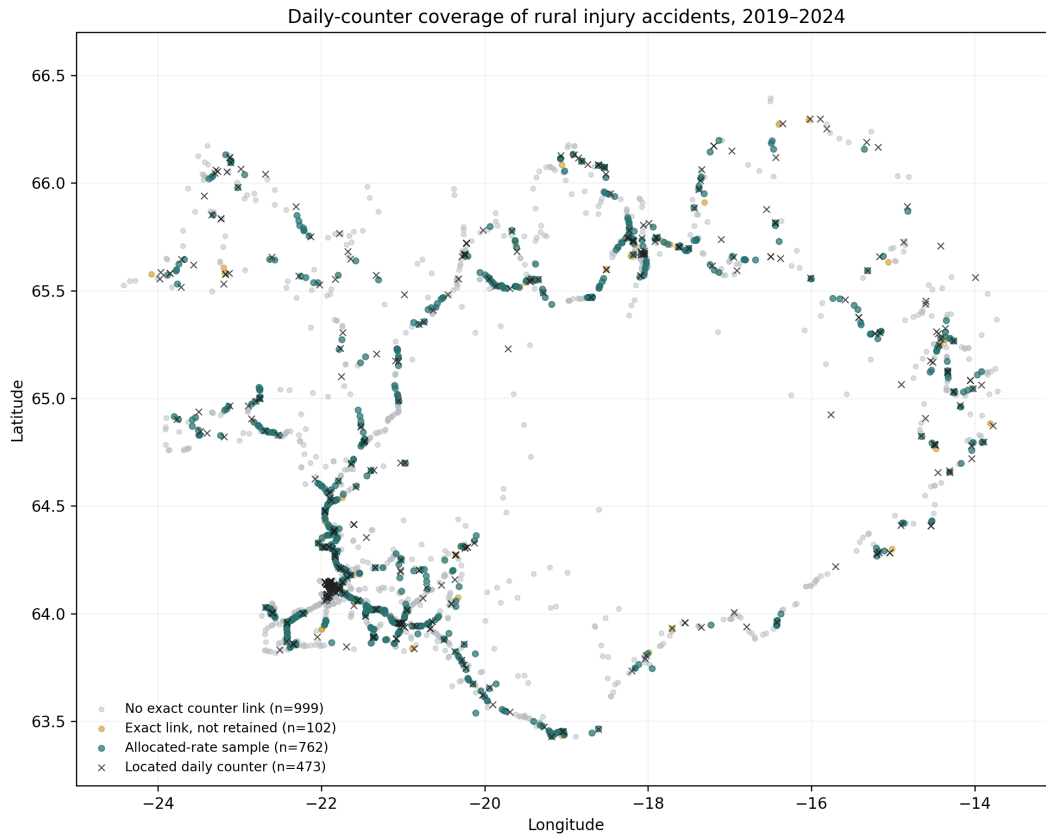


Figure A.1: Geographic coverage of located daily counters and rural injury accidents retained or excluded by the daily-counter linkage, 2019–2024.

A.3 Traffic Sensitivity Checks

The assignment-distance check shows how the broad daily-counter result changes under 5, 10, and 20 km limits. It is retained to make the spatial selection rule explicit.

Table A.4: Upper daily-counter estimate under three assignment distances

Maximum distance	Included accidents	Upper-bin accidents	Rate ratio	95% CI
5 km	585	12	2.91	1.58–5.37
10 km	750	14	2.78	1.58–4.88
20 km	767	14	2.73	1.55–4.78

Table A.5 combines the remaining traffic checks. It compares the annual model using all traffic periods with the version using only official VDU and SDU, reports the effect of excluding zero counter-days, and records exclusions caused by nonpositive VDU or derived spring/autumn daily traffic values.

Table A.5: Comparison and quality checks for the traffic analyses

Check	Data included	Estimate (95% interval)	Records
Rate model, 20-25 m/s	All VDU, SDU, and derived VHDU periods	RR 2.27 (1.68–3.07)	45 accidents
Rate model, 20-25 m/s	Official VDU and SDU periods only	RR 1.82 (1.23–2.70)	26 accidents
Rate model, ≥ 25 m/s	All VDU, SDU, and derived VHDU periods	RR 4.95 (3.07–7.98)	18 accidents
Rate model, ≥ 25 m/s	Official VDU and SDU periods only	RR 2.97 (1.47–6.02)	8 accidents
Daily traffic, 20-25 m/s	Retain zero counter-days	68.94% (62.33–74.09)	1,175 days
Daily traffic, 20-25 m/s	Exclude zero counter-days	69.29% (62.70–74.52)	1,090 days
Nonpositive published VDU	Excluded, not imputed	6.57% excluded	1,509 section-years
Nonpositive derived VHDU	Excluded, not imputed	2.40% excluded	552 section-years